

| CATEGORY          |                                                 | USB - THUMB DRIVES                                                                  |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |
|-------------------|-------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| FEATURES (30%)    | Brand                                           | Apacer                                                                              | Apacer                                                                              | Apacer                                                                              | Easydisk                                                                            | Freecom                                                                              | Nexdisk                                                                               | Princeton                                                                             | Princeton                                                                             |
|                   | Model                                           | Handy Drive 32 MB                                                                   | Handy Steno 1.1                                                                     | Handy Drive 128 MB                                                                  | 128 MB                                                                              | FM-1                                                                                 | PSD-64N                                                                               | MPDUSB64MB                                                                            | MPDUSB128MB                                                                           |
|                   | Drive Specifications                            |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |
|                   | Dimensions (LxWxH) (mm)                         | 85x28x15                                                                            | 85x28x15                                                                            | 85x28x15                                                                            | 81x22x15                                                                            | 73x22x10                                                                             | 80x20x9.5                                                                             | 59x18x10                                                                              | 59x18x10                                                                              |
|                   | Weight (in grammes)                             | 17                                                                                  | 17                                                                                  | 17                                                                                  | 11                                                                                  | 20                                                                                   | 12                                                                                    | 8                                                                                     | 8                                                                                     |
|                   | Storage capacity (in MB)                        | 32                                                                                  | 64                                                                                  | 128                                                                                 | 128                                                                                 | 32                                                                                   | 64                                                                                    | 64                                                                                    | 128                                                                                   |
|                   | Type of design (clip, keychain)                 | Keychain                                                                            | Keychain                                                                            | Keychain                                                                            | Keychain                                                                            | Keychain                                                                             | Clip                                                                                  | Keychain                                                                              | Keychain                                                                              |
|                   | Visual Indicators                               | Data transfer                                                                       | Data transfer                                                                       | Data transfer                                                                       | Power and data transfer                                                             | Power and data transfer                                                              | Power and Data transfer                                                               | Data transfer                                                                         | Data transfer                                                                         |
|                   | Interface supported                             | USB 1.1                                                                             | USB 1.1                                                                             | USB 1.1                                                                             | USB 1.1                                                                             | USB 1.1                                                                              | USB 1.1                                                                               | USB 1.1                                                                               | USB 1.1                                                                               |
|                   | Write protection button (Y/N)                   | ✓                                                                                   | ✓                                                                                   | ✓                                                                                   | ✓                                                                                   | ✗                                                                                    | ✓                                                                                     | ✗                                                                                     | ✗                                                                                     |
| PERFORMANCE (70%) | Build Quality                                   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |
|                   | Sturdiness (Scale of 5)                         | 3.5                                                                                 | 3.5                                                                                 | 3.5                                                                                 | 3                                                                                   | 4                                                                                    | 3.5                                                                                   | 3.5                                                                                   | 3.5                                                                                   |
|                   | Aesthetic design (Scale of 5)                   | 3.5                                                                                 | 3.5                                                                                 | 3.5                                                                                 | 3                                                                                   | 3                                                                                    | 4                                                                                     | 3.5                                                                                   | 3.5                                                                                   |
|                   | Package contents                                |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |
|                   | Bundled Software                                | ✗                                                                                   | ✗                                                                                   | ✗                                                                                   | FlashMail,Passid, PCLock, SecretZip                                                 | Automail, Magicdisk, PCLock, SecureZip, Password Protection utility                  | DataSheriff                                                                           | ✗                                                                                     | ✗                                                                                     |
|                   | Accessories                                     | Neckstrap, USB cable                                                                | Neckstrap, USB cable                                                                | Neckstrap, USB cable                                                                | Neckstrap, USB cable                                                                | Neckstrap, USB cable                                                                 | Neckstrap, USB cable                                                                  | ✗                                                                                     | ✗                                                                                     |
|                   | Documentation (Y/N)                             |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |
|                   | Quickstart guide                                | ✓                                                                                   | ✓                                                                                   | ✓                                                                                   | ✓                                                                                   | ✓                                                                                    | ✓                                                                                     | ✗                                                                                     | ✗                                                                                     |
|                   | Manual                                          | on CD                                                                               | on CD                                                                               | on CD                                                                               | on CD                                                                               | ✗                                                                                    | on CD                                                                                 | ✗                                                                                     | ✗                                                                                     |
|                   | Sisoft Sandra 2003 Professional Drive Benchmark |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |
| OVERALL SCORE     | Drive Index                                     | 907                                                                                 | 868                                                                                 | 919                                                                                 | 877                                                                                 | 686                                                                                  | 674                                                                                   | 945                                                                                   | 941                                                                                   |
|                   | Sequential Read (KBps)                          | 960                                                                                 | 928                                                                                 | 973                                                                                 | 926                                                                                 | 892                                                                                  | 887                                                                                   | 970                                                                                   | 988                                                                                   |
|                   | Random Read (KBps)                              | 958                                                                                 | 925                                                                                 | 967                                                                                 | 924                                                                                 | 508                                                                                  | 477                                                                                   | 967                                                                                   | 982                                                                                   |
|                   | Sequential Write (KBps)                         | 698                                                                                 | 637                                                                                 | 714                                                                                 | 686                                                                                 | 539                                                                                  | 564                                                                                   | 853                                                                                   | 762                                                                                   |
|                   | Random Write (KBps)                             | 700                                                                                 | 628                                                                                 | 716                                                                                 | 680                                                                                 | 378                                                                                  | 360                                                                                   | 850                                                                                   | 766                                                                                   |
|                   | HDTach                                          |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |
|                   | Random Access Time (in msec)                    | 6                                                                                   | 6                                                                                   | 6                                                                                   | 6                                                                                   | 6                                                                                    | 68.5                                                                                  | 6                                                                                     | 6                                                                                     |
|                   | Average read speed (MBps)                       | 1                                                                                   | 1                                                                                   | 1                                                                                   | 1                                                                                   | 1                                                                                    | 1                                                                                     | 1                                                                                     | 1                                                                                     |
|                   | Average write speed (MBps)                      | 1                                                                                   | 1                                                                                   | 1                                                                                   | 1                                                                                   | 0.1                                                                                  | 0.1                                                                                   | 1                                                                                     | 1                                                                                     |
|                   | Read Burst speed (MBps)                         | 1                                                                                   | 1                                                                                   | 1                                                                                   | 1                                                                                   | 0.9                                                                                  | 0.9                                                                                   | 1                                                                                     | 1                                                                                     |
| OVERALL SCORE     | CPU utilisation (%age)                          | 0.1                                                                                 | 0                                                                                   | 0.1                                                                                 | 0                                                                                   | 0                                                                                    | 0                                                                                     | 0                                                                                     | 0                                                                                     |
|                   | Real-world tests: File Transfer Tests (seconds) |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |
|                   | Sequential Read (20 MBps)                       | 4.2                                                                                 | 7.57                                                                                | 7.14                                                                                | 9.27                                                                                | 8.15                                                                                 | 6.34                                                                                  | 9.67                                                                                  | 8.3                                                                                   |
|                   | Assorted Read (20 MBps)                         | 25.35                                                                               | 24.17                                                                               | 34.37                                                                               | 24.67                                                                               | 43.69                                                                                | 24.67                                                                                 | 23.54                                                                                 | 23.45                                                                                 |
|                   | Sequential Write (20 MBpsec)                    | 34.47                                                                               | 33.87                                                                               | 30.66                                                                               | 36.16                                                                               | 40.29                                                                                | 36.16                                                                                 | 29.01                                                                                 | 30.66                                                                                 |
|                   | Assorted Write (20 MBps)                        | 65.64                                                                               | 63.58                                                                               | 55.21                                                                               | 55.29                                                                               | 117.29                                                                               | 55.29                                                                                 | 44.17                                                                                 | 55.21                                                                                 |
|                   | Maximum capacity transfer test (in MBps)        |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |
|                   | Sequential Read (Max capacity) (MBps)           | 1.34                                                                                | 1.28                                                                                | 0.93                                                                                | 1.56                                                                                | 1.20                                                                                 | 1.21                                                                                  | 1.41                                                                                  | 2.30                                                                                  |
|                   | Assorted Read (Max capacity) (MBps)             | 1.13                                                                                | 1.72                                                                                | 1.23                                                                                | 1.18                                                                                | 0.73                                                                                 | 1.09                                                                                  | 1.77                                                                                  | 1.26                                                                                  |
|                   | Sequential Write (Max Capacity) (MBps)          | 0.65                                                                                | 0.61                                                                                | 0.69                                                                                | 0.71                                                                                | 0.51                                                                                 | 0.53                                                                                  | 0.72                                                                                  | 0.74                                                                                  |
| OVERALL SCORE     | Assorted Write (Max capacity) (MBps)            | 0.52                                                                                | 0.43                                                                                | 0.57                                                                                | 0.52                                                                                | 0.36                                                                                 | 0.38                                                                                  | 0.57                                                                                  | 0.62                                                                                  |
|                   | Cost per MB                                     | 50.00                                                                               | 36.69                                                                               | 29.44                                                                               | 28.69                                                                               | 72.58                                                                                | 44.35                                                                                 | 40.32                                                                                 | 30.24                                                                                 |
|                   | Features                                        | 20.5                                                                                | 20.5                                                                                | 10.5                                                                                | 12                                                                                  | 8.6                                                                                  | 22.7                                                                                  | 3.5                                                                                   | 3.5                                                                                   |
|                   | Performance                                     | 45.17                                                                               | 44.23                                                                               | 55.56                                                                               | 55.61                                                                               | 40.16                                                                                | 34.94                                                                                 | 49.33                                                                                 | 47.65                                                                                 |
| OVERALL SCORE     | Overall Score (Performance+Features)            | 65.67                                                                               | 64.73                                                                               | 66.06                                                                               | 67.61                                                                               | 48.76                                                                                | 57.64                                                                                 | 52.83                                                                                 | 51.15                                                                                 |
|                   | Value for Money (in Rs)                         | 1.31                                                                                | 1.76                                                                                | 2.24                                                                                | 2.36                                                                                | 0.67                                                                                 | 1.30                                                                                  | 1.31                                                                                  | 1.69                                                                                  |
|                   | Grade                                           |  |  |  |  |  |  |  |  |
|                   | Price (Rs)                                      | 1,550                                                                               | 2,275                                                                               | 3,650                                                                               | 3,500                                                                               | 2,250                                                                                | 2,750                                                                                 | 2,500                                                                                 | 3,750                                                                                 |

Value for money

Unlike the USB thumb drives comparison, where the actual cost per MB was very high, the large storage capacities of the external hard drive ensure that the cost per MB stays low. While evaluating the Value for Money quotient, we didn't just consider the cost per MB, but also took into account the general performance of each thumb drive.

The Iomega drives sped ahead of the others, powered by their highly economical costs of 13 paise and 14 paise per MB for the 120 GB and 80 GB drives, respectively. They also scored

respectably in the benchmarks and tests. In the smaller 4,200-rpm drive VFM-battle, the most cost-effective drive was the Transcend 40 GB drive—not suprising since it cost only 29 paise for MB.

Conclusion

Finally, after all our tests, we saw that although the so-called desktop hard drives had a seemingly unfair advantage with their greater speeds, the slower drives still managed to perform well in the real-world tests. The eventual winner was the 7,200-